

Perioperative Transcutaneous Electrical Acupoint Stimulation for Postoperative Pain Relief Following Laparoscopic Surgery

A Randomized Controlled Trial

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Objectives: This trial was conducted to assess the influence of transcutaneous electrical acupoint stimulation (TEAS) on postoperative pain intensity and the optimal time of TEAS application during perioperative period in patients undergoing laparoscopic surgery.

Methods: From July 2012 to October 2013, 380 patients scheduled for laparoscopic surgery under general anesthesia were randomly assigned to receive sham TEAS (group SSS), preoperative TEAS (group TSS), preoperative TEAS combined with intraoperative TEAS (group TTS) or preoperative TEAS combined with postoperative TEAS (group TST) ($n = 95$ each group). Primary outcomes included resting and activity pain intensity evaluated by visual analog scale at 1, 6, 24, and 48 hours after surgery.

Results: At postoperative 6 hours, the activity pain intensity was significantly lower in groups TTS and TST compared with groups SSS and TSS ($P < 0.001$). At postoperative 24 and 48 hours, activity pain intensity decreased in group TST compared with both groups SSS and TSS ($P < 0.001$). The supplemental analgesic requirement was less in group TST compared with group SSS. There was no significant difference in intraoperative anesthetic consumption, postoperative nausea and vomiting and time of the first postoperative flatus or defecation among the 4 groups. The patient satisfactory rate was higher in groups TSS, TTS, and TST compared with group SSS. No side effect related to TEAS was observed during the postoperative 48 hours.

Discussion: Combination of preoperative TEAS with intraoperative or postoperative TEAS, rather than preoperative TEAS alone, is an effective and safe adjunctive for management of postoperative pain following laparoscopic surgery.

Key Words: postoperative pain, transcutaneous electrical acupoint stimulation, laparoscopic surgery, visual analog scale

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Optimal management of acute postoperative pain is a key component of enhanced recovery after surgery, particularly given its adverse implications on patients' physical activities, recovery time, quality of life, morbidity, and health care costs.¹ However, it is well documented that postoperative pain management often remains insufficient, and there is no international consensus about the optimal analgesic intervention after most surgical procedures, including laparoscopic surgery.^{2–4} Laparoscopic surgery has advantages over traditional open surgery in respect of postoperative pain, shortened hospital stay, and enhanced postoperative recovery. Nevertheless, laparoscopic surgery still results in substantial postoperative pain around the incision site.^{5,6} It is reported that only 60% of patients are satisfied with postoperative pain relief⁷ and about 30% to 40% of discharged patients experience moderate to severe pain after laparoscopic surgery.⁸ The use of more than 1 analgesic modality (multimodal analgesia) to achieve effective pain control while minimizing the side effects of opioids that delay discharge for patients undergoing laparoscopic procedure has been required in clinical practice.

Transcutaneous electrical acupoint stimulation (TEAS), an acupuncture-related technique, is a form of noninvasive electrical stimulation that produces a perceptible sensation through electrodes placed on acupoints. Compared with conventional acupuncture, it has no risk of infections and can potentially be applied by medical personnel with minimal training.^{9,10} It has been shown that TEAS reduces the consumption of intraoperative anesthetics, reduces the incidence of general anesthesia-related side effects and improves the quality of postoperative recovery.^{11–13} Furthermore, TEAS can reduce postoperative pain intensity and perioperative analgesic requirements in colonoscopy,¹⁴ lumbar spine surgery,¹⁵ ambulatory breast surgery,¹² supratentorial craniotomy,¹⁶ and video-assisted thoracic surgical lobectomy.¹³ However, there is no study having assessed the effects of TEAS performed at different perioperative timepoints on postoperative pain intensity and the optimal time of TEAS application during perioperative period in patients undergoing laparoscopic surgery.

We hypothesized that TEAS could reduce postoperative pain following laparoscopic surgery and time of TEAS application during perioperative period could affect its postoperative analgesic effects.

MATERIALS AND METHODS

After the study was approved by the Institutional Review Board and the protocol was registered in ClinicalTrials.gov

(ChiCTR-IOR-15006032), this randomized controlled trial was performed from July 2012 to October 2013 at the second affiliated hospital of Zhejiang University, China. The study included patients who were 18 to 70 years old, American Society of Anesthesiologists class I or II and scheduled for laparoscopic cholecystectomy or laparoscopic gynecological surgery. Exclusion criteria included implantation of cardiac pacemaker, history of using acupuncture or acupuncture-related techniques, history of chronic pain or daily use of analgesics, history of excess or long-term alcohol use, inability to understand the consent form or how to use a visual analog scale (VAS) for pain measurement, pregnancy, renal or hepatic insufficiency, history of asthma or gastrointestinal ulcer or hemorrhage tendency or other contradictions of flurbiprofenaxetil. Patients were withdrawn from the study if surgery time exceeded 200 minutes, or laparoscopic procedure was converted to open procedure, or intraoperative complications happened, such as intraoperative bleeding requiring blood transfusion.

All participants followed a standard perioperative care protocol. After the patient arrived at the preoperative preparation room, an investigator explained the details of the study protocol to the recruited patient and informed him/her that he/she had the right to decline participation. After the written informed consent was obtained, anesthetic evaluation was performed by the anesthesia team, who were not aware of patient's group assignment. Then a SPSS-generated random number table was used to allocate the patients into 1 of the 4 groups in a 1:1:1:1 ratio. Group SSS received sham TEAS preoperatively, sham TEAS intraoperatively, and sham TEAS postoperatively; group TSS received true TEAS preoperatively, sham TEAS intraoperatively, and sham TEAS postoperatively; group TTS received true TEAS preoperatively, true TEAS intraoperatively, and sham TEAS postoperatively; group TST received true TEAS preoperatively, sham TEAS intraoperatively, and true TEAS postoperatively. The TEAS operators were informed the group allocation of the patient by a sealed opaque envelope and they were the only individuals aware of the treatment allocation, while patients, anesthesiologists and evaluators were blinded to the treatment allocation.

Before the intervention, acupoints named Hegu (LI4) and Neiguan (P6) were identified and selected in this study because these were one of the traditional acupuncture points used in relieving pain and related side effects¹⁷ that would not interfere with laparoscopic practice. Hegu (LI4) is located at the midpoint of the dorsum of the second metacarpus on the radial side, and Neiguan (P6) is located 5 cm above the transverse crease of the wrist between the tendons of palmaris longus and flexor carpi radialis.^{10,17} TEAS was achieved by a battery-powered unit named HANS Acupoint Nerve Stimulator (HANS-200A, Nanjing Jisheng Medical Technology Company, Nanjing, China), which sent electrical impulses through 2 pairs of rubber electrodes (4×4 cm) attached to participant's bilateral acupoints. Patients in the TEAS treatment groups received an alternating dense and disperse stimulation, where the 2 Hz (0.6 ms pulse width) stimulation is alternated with 100 Hz stimulation (0.2 ms pulse width) automatically in every 3 seconds,^{9,13,18} at a maximum current tolerated but subnoxious to the patient.¹⁹ Each treatment lasted 30 minutes in each perioperative period. For sham TEAS, patients were stimulated with the minimal current that the patient could feel, with the same alternating frequency as

described above, for 5 minutes, and the electrodes were applied to the same acupoints as in the study group and were retained in situ for 30 minutes.

During the preoperative evaluation, patients were instructed to use a 10-cm pain VAS (0 = no pain, 10 = worst imaginable pain). No premedication was administered. During the operation, standardized monitoring and bispectral index were applied. Anesthesia was induced with propofol 2.0 to 4.0 mg/kg and sufentanil 0.5 to 1.0 µg/kg, and rocuronium 0.6 mg/kg was administered to provide neuromuscular blockade for tracheal intubation. Anesthesia was maintained with adjusted propofol and remifentanyl infusion to maintain the bispectral index 40 to 50. Propofol and remifentanyl infusion were continued until skin suturing was completed. To evaluate the anesthetic consumption, opioids were converted to morphine using the following equivalencies: 0.01 mg of sufentanil = 0.1 mg of remifentanyl = 10 mg of morphine.²⁰

After surgery, patients were transferred intubated to the postanesthesia care unit (PACU) where they were consecutively monitored. Each case was extubated and discharged from PACU if meeting the clinical criteria.²¹ No local anesthetic was used before skin incision or after conclusion of the procedure. According to our clinical routine postoperative analgesia scheme, all patients received intravenous flurbiprofenaxetil 50 mg at the end of surgery, and intravenous flurbiprofenaxetil 50 mg was supplemented if the analgesia was inadequate (VAS ≥ 4) or if patients required analgesics within 48 hours. All patients were also given tropisetron 5 mg at the end of surgery, and additional 5 mg was given intravenously as rescue, if vomiting occurred or if persistent nausea was reported for 2 hours.

The primary outcomes were resting and activity pain intensity evaluated by VAS at 1, 6, 24, and 48 hours after surgery. The activity pain intensity was measured by VAS as the most severe pain experienced during coughing. Secondary outcomes included intraoperative propofol and opioid consumption, incidence of postoperative nausea and vomiting, supplementary analgesic and antiemetic requirements within 48 hours, extubation time, length of stay in PACU, time of the first postoperative flatus and defecation, and patient satisfaction. Patient satisfaction was evaluated with a 5-point scale (1 = unsatisfactory, 5 = extremely satisfactory), and the incidence of the "extremely satisfactory" report defined the satisfactory rate. Potential side effects of TEAS, such as local irritation, redness, contact dermatitis, or muscle ache, were recorded.

The sample size was calculated based on a pilot study developed with 20 patients undergoing laparoscopic surgery, using a 2-tailed power analysis with 0.90 power and an α of 0.05. Assuming a VAS of 4 in the control group, it should decrease by 25% with perioperative TEAS, which is defined as the minimally important difference in pain.^{22,23} With an estimated SD of 2, 77 patients were needed per study group. Allowing for a possible dropout, 95 patients per group were included in the study so as to ensure enough data to fit the analysis of variance models and to allow for comparisons among other outcome variables of interest.

Statistical analysis of data were done using PRISM (version 5.01; GraphPad Software, San Diego, CA). All data were tested for normal distribution using the Shapiro-Wilk test. Normally distributed data (including baseline characteristics and some of the secondary outcomes) were analyzed with 1-way analysis of variance are expressed as

mean $\pm$ SD. Non-normally distributed data (including pain intensity at each assessment timepoint) were analyzed with Kruskal-Wallis test for analysis between groups followed by Dunn's multiple comparison tests, and are expressed as median (interquartile range). Two-way analysis of variance with Bonferroni post hoc test was used to evaluate the effect of the intervention (group SSS, group TSS, group TTS, and group TST) and the assessment timepoint. Categorical variables were analyzed by χ^2 test or Fisher test as appropriate and are reported as percentage (95% confidential interval). Bonferroni correction was applied. $P < 0.05$ was considered statistically significant.

RESULTS

During the study period, 435 patients were assessed for eligibility. After exclusion of 55 patients, 380 patients were randomized into 4 groups with 95 patients per group. A total of 19 patients were withdrawn by the end of the trial due to follow-up lost ($n = 8$), change of surgery ($n = 6$), prolonged surgery time ($n = 2$), and intraoperative complications ($n = 3$), leaving 361 patients for analysis (Fig. 1). The average age of the study population was 44.9 ± 11.6 years and 82.3% of study participants were female. One hundred and seventy-six (48.8%) patients underwent laparoscopic cholecystectomy and 185 (51.2%) underwent laparoscopic gynecological surgery. There was no significant difference among the 4 groups in demography and intraoperative characteristics, including age, sex, weight, body mass index, American Society of Anesthesiologists classification, types of surgery, anesthesia duration, or surgery duration ($P > 0.05$) (Table 1).

There was no significant difference among the 4 groups in the resting pain intensity at 1, 6, 24, and 48 hours after surgery (Fig. 2). In 2-way analysis of variance, group TSS had no impact on activity pain intensity compared with group SSS at each timepoint as evaluated ($P > 0.05$, Fig. 3).

However, groups TTS and TST mitigated activity pain compared with groups SSS and TSS ($P < 0.001$). At postoperative 1 hour, the difference of activity pain intensity between the 4 groups was not significant as shown in Figure 3. However, the activity pain intensity at postoperative 6 hours was significantly lower in groups TTS [3.0 (3.0 to 4.0)] and TST [3.0 (2.5 to 5.0)] compared with groups SSS [5.0 (4.0 to 6.0)] and TSS [5.0 (5.0 to 6.0)] ($P < 0.001$). The activity pain intensity at postoperative 24 hours significantly decreased in group TST [3.0 (3.0 to 4.0)] compared with groups SSS [4.0 (3.8 to 5.0)] and TSS [4.0 (4.0 to 5.0)] ($P < 0.001$), and in group TTS [4.0 (3.0 to 4.0)] compared with group TSS [4.0 (4.0 to 5.0)] ($P < 0.001$). The activity pain intensity at postoperative 48 hours was significantly reduced in group TST [3.0 (2.0 to 3.0)] compared with groups SSS [3.0 (3.0 to 4.0)] and TSS [3.0 (3.0 to 4.0)] ($P < 0.001$). There was no significant differences between groups TTS and TST at any timepoint ($P > 0.05$) (Fig. 3).

None of the study participants required more than 1 supplemental analgesic. The supplemental analgesic requirements within the first postoperative 48 hours were less in group TST compared with group SSS (7.9% vs. 23.3%, $P = 0.020$). Although groups TSS and TTS required less supplemental analgesics than group SSS, the difference was not significant ($P > 0.05$) (Table 2).

There was no significant difference in intraoperative propofol and opioid consumption among the 4 groups (Table 2). The incidence of nausea, vomiting and additional antiemetic requirements within the first postoperative 48 hours were 30.2%, 25.8%, and 26.0%, respectively. No differences existed between groups (Table 2). There was no significant difference in time of the first postoperative flatus or defecation, extubation time, and length of stay in PACU among the 4 groups. The patient satisfactory rate was higher in groups TSS, TTS, and TST compared with group SSS (Table 2). No side effect related to TEAS was observed during the postoperative 48 hours.

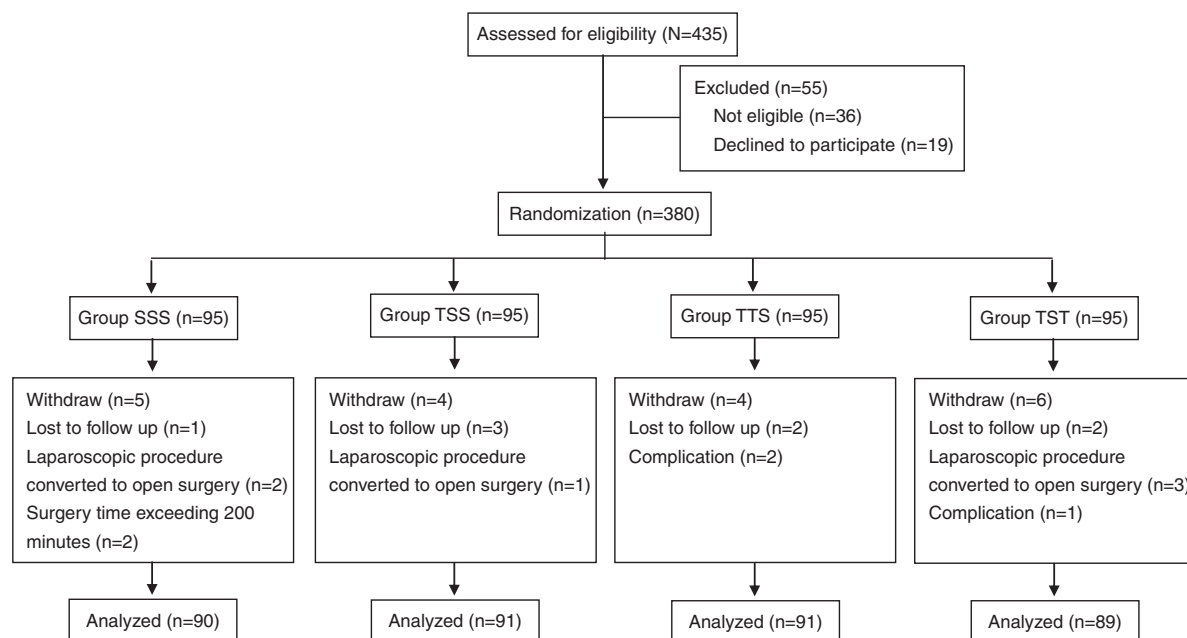


FIGURE 1. Flowsheet.

TABLE 1. Demographic and Intraoperative Characteristics of Patients in the 4 Groups

Outcomes	Group SSS (n = 90)	Group TSS (n = 91)	Group TTS (n = 91)	Group TST (n = 89)
Age (y)	46.5 ± 13.6	45.9 ± 10.2	42.4 ± 10.4	44.8 ± 11.5
Sex (Male, %)	20.0 (11.7-28.3)	14.3 (7.1-21.5)	16.5 (8.8-24.2)	20.2 (11.8-28.6)
Weight (kg)	59.3 ± 10.5	60.0 ± 9.2	57.8 ± 8.4	60.1 ± 10.2
Body mass index (kg/m ²)	22.7 ± 3.1	22.8 ± 2.3	22.6 ± 2.8	23.3 ± 3.2
ASA (I, %)	31.1 (21.5-40.7)	36.3 (26.4-46.2)	23.1 (14.4-31.8)	32.6 (22.8-42.4)
Type of surgery (laparoscopic cholecystectomy, %)	53.3 (42.9-63.7)	49.5 (39.2-59.8)	41.8 (31.6-52.0)	50.6 (40.2-61.0)
Time of anesthesia (min)	90.6 ± 26.7	102.1 ± 36.0	97.5 ± 32.1	97.3 ± 34.1
Time of surgery (min)	66.3 ± 27.8	77.1 ± 34.4	74.7 ± 30.1	73.2 ± 32.5

Continuous variables are presented as mean ± SD and categorical variables are presented as percentage (95% confidential interval). ASA indicates American Society of Anesthesiologists.

DISCUSSION

In this randomized, patient and anesthesiologist evaluator blinded, controlled clinical trial, preoperative TEAS alone was ineffective in reduction of postoperative pain. However, combined preoperative TEAS with intraoperative

or postoperative TEAS reduced postoperative pain within the first 48 hours, as demonstrated by a significant reduction of activity pain intensity and improved patient satisfaction. Furthermore, combined preoperative TEAS with postoperative TEAS was associated with a concomitantly

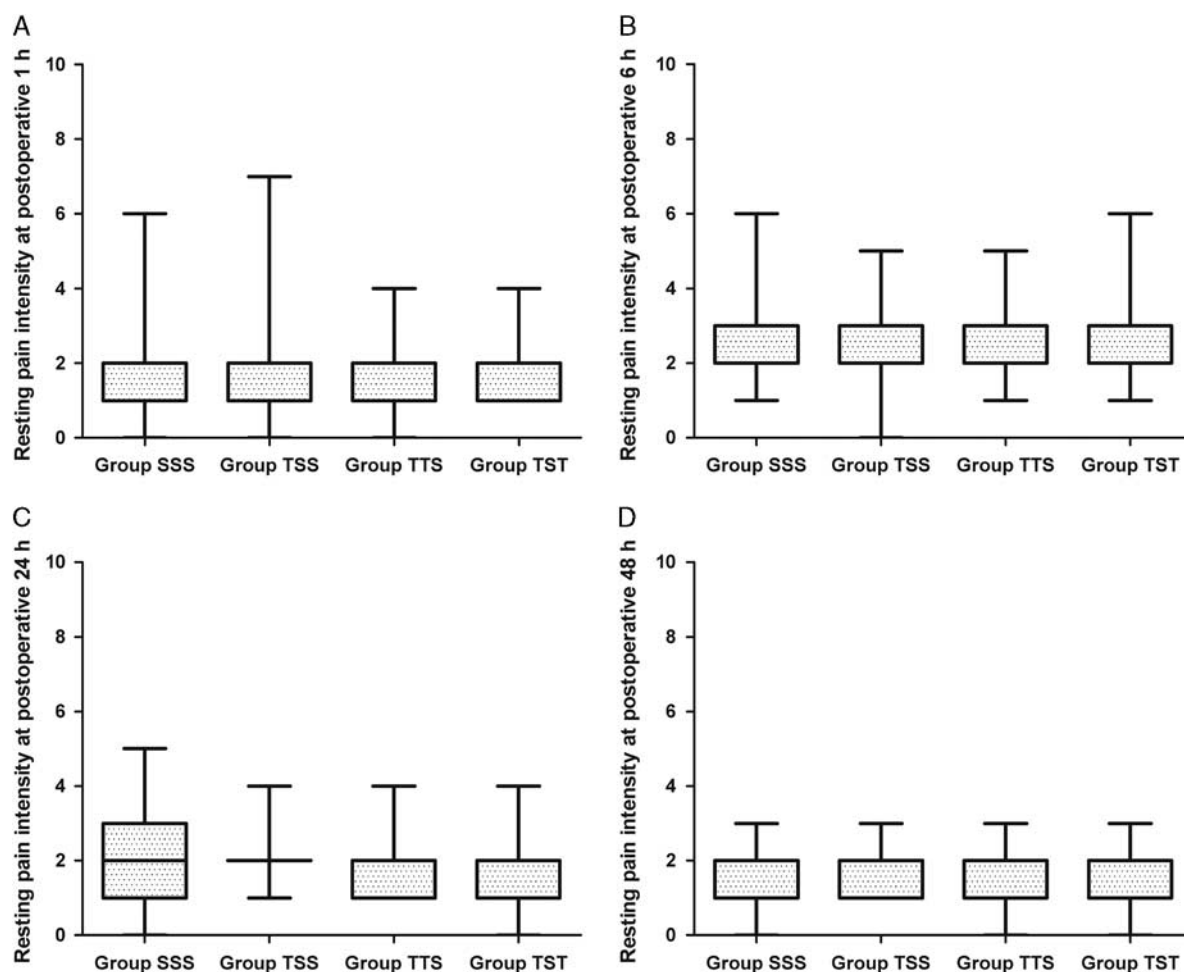


FIGURE 2. The dynamic changes of resting pain intensity evaluated by visual analog scale at postoperative 1 (A), 6 (B), 24 (C), and 48 hours (D). Two-way analysis of variance with Bonferroni post hoc test was used to evaluate the effect of the groups and the assessment timepoint. For each assessment timepoint, Kruskal-Wallis test followed by Dunn's multiple comparison tests was used to compare differences among groups. Boxes represent first/third quartile, lines represent medians, and whiskers represent the minimum and maximum values.

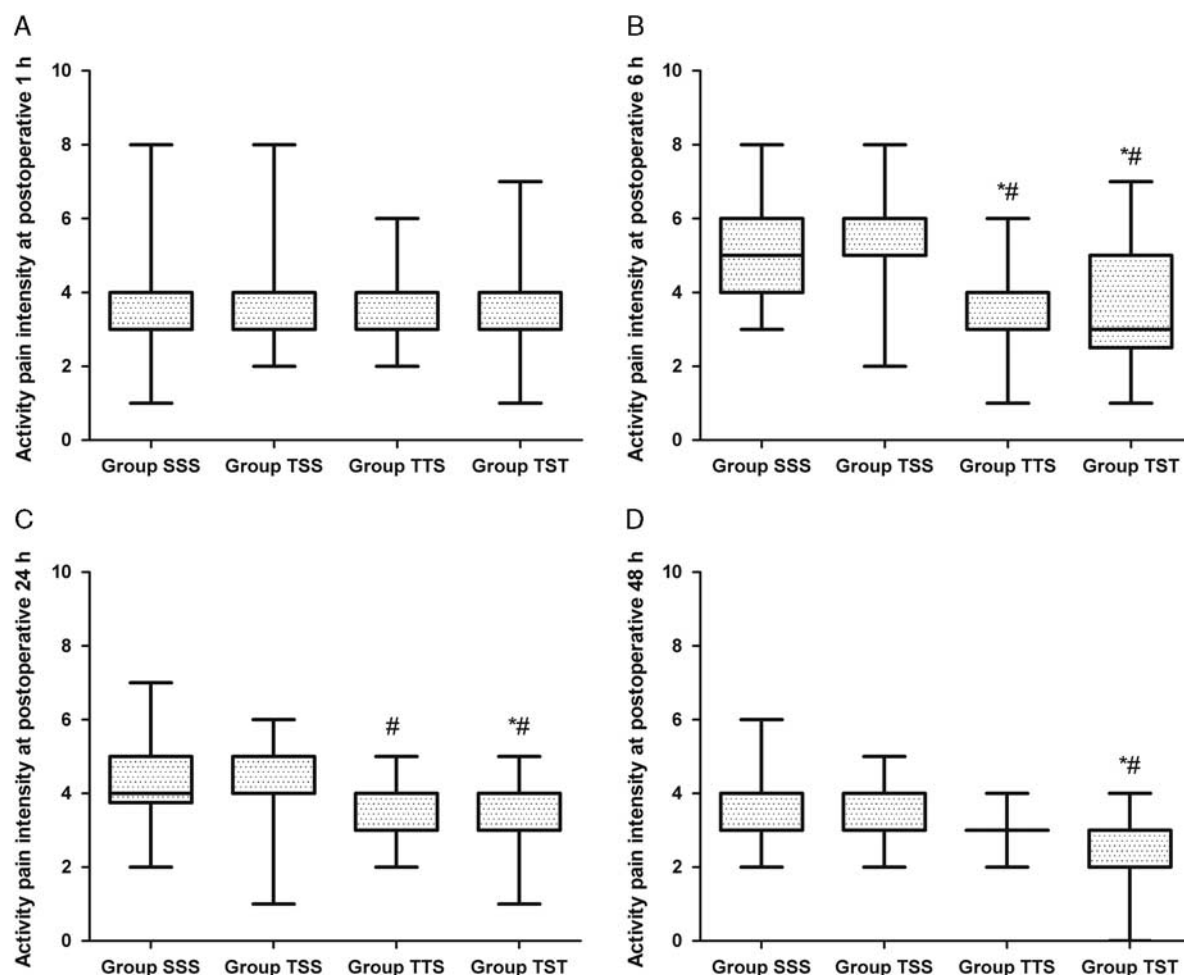


FIGURE 3. The dynamic changes of pain intensity evaluated by activity visual analog scale (VAS) at postoperative 1 (A), 6 (B), 24 (C) and 48 hours (D). Two-way analysis of variance with Bonferroni post hoc test was used to evaluate the effect of the groups and the assessment timepoint. For each assessment timepoint, Kruskal-Wallis test followed by Dunn's multiple comparison tests was used to compare differences among groups. *Compared with group SSS, $P < 0.05$. #Compared with group TSS, $P < 0.05$. Boxes represent first/third quartile, lines represent medians, and whiskers represent the minimum and maximum values.

TABLE 2. Comparison of Secondary Outcomes Among the 4 Groups

Outcomes	Group SSS (n = 90)	Group TSS (n = 91)	Group TTS (n = 91)	Group TST (n = 89)	P
Supplemental analgesics (%)	23.3 (14.5-32.1)	11.0 (4.5-17.5)	15.4 (8.0-22.8)	7.9 (2.2-13.58)*	0.020
Intraoperative propofol (mg/kg)	9.3 ± 4.4	10.0 ± 4.8	10.5 ± 3.4	10.0 ± 4.0	0.340
Intraoperative morphine (mg/kg)	1.5 ± 0.4	1.6 ± 0.4	1.6 ± 0.4	1.5 ± 0.4	0.303
Nausea (%)	25.6 (16.5-34.6)	34.1 (24.3-43.9)	26.4 (17.3-35.5)	34.8 (24.8-44.8)	0.376
Vomiting (%)	22.2 (13.6-30.8)	25.3 (16.3-34.3)	26.4 (17.3-35.5)	29.2 (19.7-38.7)	0.759
Additional antiemetic (%)	23.3 (14.5-32.1)	25.3 (16.3-34.3)	26.4 (17.3-35.5)	29.2 (19.7-38.7)	0.840
Extubation time (min)	22.5 ± 13.2	20.8 ± 15.8	18.4 ± 10.1	21.5 ± 12.3	0.180
Length of stay in PACU (min)	66.0 ± 16.0	62.2 ± 19.4	62.7 ± 16.3	64.8 ± 15.5	0.379
First postoperative passed flatus (h)	41.5 ± 19.6	45.2 ± 19.9	44.7 ± 16.6	41.6 ± 18.1	0.391
First postoperative defecation (h)	63.4 ± 22.9	63.6 ± 19.8	61.8 ± 15.7	57.3 ± 20.6	0.127
Satisfaction (%)	88.9 (82.4-95.4)	98.9 (96.7-101.1)*	100 (100-100)*	100 (100-100)*	< 0.001

Continuous variables are presented as mean ± SD and categorical variables are presented as percentage (95% confidential interval).

*Compared with group SSS, $P < 0.05$.

PACU indicates postanesthesia care unit.

decreased analgesic intake. Thus, this study confirms our hypothesis that time of TEAS application during perioperative period could affect its postoperative analgesic effects following laparoscopic surgery.

Early postoperative mobilization is essential for rapid recovery, but a prerequisite for early postoperative mobilization was effective analgesia.²⁴ Contemporary analgesic techniques available for multimodal postoperative analgesia include pharmacological and nonpharmacological techniques. However, both systemic and local pharmacological modalities have their own advantages and specific safety profile, which highlights the importance of selecting the adjunctive nonpharmacological techniques.²⁵ Adjunctive nonpharmacological techniques include acupuncture and related techniques, music therapy, and hypnosis, among others. A lack of harm is associated with such techniques, but there is mixed evidence about the effectiveness. Music therapy and hypnosis may decrease perception of pain through mechanisms such as attention shift or cognitive coping, but the effect is transitory and not all patients may respond similarly.²⁶ Our work was the first to find that combined preoperative TEAS with intraoperative or postoperative TEAS in patients undergoing laparoscopic surgery was superior to sham TEAS or preoperative TEAS alone as a complementary nonpharmacologic therapy for postoperative pain relief. Advantages of TEAS include noninvasiveness, easy application and absence of complications, adverse effects, and interference with patients' consciousness or mobility.¹⁰ The successful integration of the TEAS into laparoscopic surgery stands as an example of an innovative and feasible approach to multimodal pain management, which is exciting and encouraging for patient recovery.

In this study, perioperative application of repeated TEAS as a supplement to pharmacologic analgesia lowered activity pain intensity, but not rest pain intensity. This finding is consistent with meta-analyses of past research showing TEAS reduces pain intensity beyond that of pharmacologic analgesia.²⁷ It is also supportive of a past trial suggesting transcutaneous electrical nerve stimulation reduced pain intensity during walking and deep breathing, but had no effect on pain at rest.²⁸ The possible mechanism was that peripheral electrical stimulation reduces primary mechanical hyperalgesia and that a reduction in hyperalgesia corresponds to a reduction in activity-evoked pain.^{28,29} In contrast, pain at rest might be influenced by different mechanisms than pain with movement and a reduction in hyperalgesia does not correspond to a decrease in pain at rest.

Because of the residuary effect of intraoperative opioids and analgesics administered at the end of surgery, activity pain intensity at postoperative 1 hour was low and the effect of TEAS might be covered. At postoperative 6 hour, activity pain intensity was stable or increased to some degree in groups SSS and TSS. In contrast, activity pain intensity significantly decreased in groups TTS and TST, indicating that preoperative TEAS combined with intraoperative or postoperative TEAS provided an analgesic effect, which lasted at least 6 hour. Importantly, as VAS of 4 was considered as the threshold for intervention in clinical practice, our discovery that the combined preoperative TEAS with intraoperative or postoperative TEAS reduced the medians of activity pain intensity from moderate range (4 or 5) to the mild range (3) at postoperative hour 6 and/or 24 was significantly meaningful for patients with obstructive

sleep apnea, whom may be prone to complications (primarily respiratory depression) from analgesics, and patients with contradictions for analgesics. Besides, satisfactory rate in treatment groups increased by approximately 10% compared with sham TEAS group. In other words, appropriate application of TEAS not only decreased postoperative pain intensity and analgesic requirement, but also improved patient experience, as highlighted by patient-centered care of the surgical patient and perioperative surgical home.^{30,31}

The analgesic effects of TEAS are achieved mainly by preventing pain messages from ascending in the spinothalamic tract by increasing A- β fiber transmission³² and by stimulation of the release of central endogenous opioid peptides.^{18,33} Differential release of opioid peptides in the central nervous system by TEAS has been noted, with a frequency of 2 Hz triggering the release of enkephalins and β -endorphins, and 100 Hz stimulation selectively increasing the release of dynorphin in the spinal cord. A combination of both frequencies as applied in our study allows synergistic interaction among the three endogenous opioid peptides and provides a relatively powerful analgesic effect.^{9,18} In this study, a maximum current tolerated by the patients was chosen in treatment groups. The effects of intensity of electroacupuncture upon experimental pain was investigated by Barlas et al,¹⁹ whose study found that high intensity levels may be the optimal dose selection. And thereafter, clinical trials focusing on acupuncture or acupuncture-related techniques such as TEAS, followed their protocol, which defined the optimal intensity as "to tolerance but subnoxious" exactly.^{19,34–36}

Some research have focused on the impact of acupuncture and related techniques on postoperative pain after laparoscopic surgery. However, the conclusions are conflicting. Some support the perioperative application of acupuncture, whereas others failed.^{36–40} This leads to confusion in clinical practice. We consider that the previous conflicting results should be interpreted as to the strength of the study protocol, types of surgery, and acupuncture characteristics, such as type (needling with or without electrical stimulation, acupressure, TEAS), electrode position, duration, frequency, intensity and sessions of stimulation. Yao et al³⁶ showed that TEAS (bilateral Hegu, Neiguan, Zusanli, and Sanyinjiao) 30 minutes before surgery was effective in improving the quality of recovery, postoperative analgesia and patient satisfaction, while Ng et al's^{37,38} data underlined the importance of postoperative electroacupuncture. In contrast, intraoperative auricular acupuncture has not been shown to change postoperative analgesic requirement.^{39,40} Given these conflicting results, the 4 different groups in this study were designed to evaluate effects of TEAS application times during the perioperative period on postoperative pain and analgesic requirements.

To better understand the results, possible mechanisms of different effects in groups TSS, TTS and TST should be demonstrated. The present study showed the analgesic effect of a single session of 30-min preoperative TEAS was not significant as evaluated after the 1 to 2 hours laparoscopic procedure, which was consistent with the previous finding that the effect of a single session of electroacupuncture lasted only approximately 2 hours.⁴¹ It has been reported that a period of 30 to 40 minutes is required for the full expression of analgesic effect in normal volunteers.^{42,43} A shorter time such as 5 or 20 minutes is not the optimal choice,^{42,44} and more than 1 to 2 hours may lead to

a diminution of analgesic effect, a phenomenon known as “acupuncture tolerance.”^{43,45} Therefore, excessive prolongation of TEAS duration is not recommended. To improve the analgesic effect of TEAS, we set 2 more groups receiving 2 sessions of stimulation perioperatively. Interestingly, 2 separate sessions of TEAS effectively relieved postoperative pain for up to 48 hours. The possible explanatory mechanisms were that repeated stimulation increased the amount of endogenous opioids released, which could in turn result in a much more significant analgesia effect and longer period of pain relief. And the sustained effect might be due to the induced expression of preproenkephalin messenger ribonucleic acid (mRNA) and preprodynorphin mRNA as demonstrated in rat brain,⁴⁶ and the genetic expression of neurotransmitters prolonged the effect.

The major strengths of this study are a large sample size and randomized, anesthesiologist patient and evaluator blinding and sham TEAS-controlled design. After randomization, both the preoperative and intraoperative characteristics were not significantly different among study groups, eliminating the potential influences of these factors on postoperative pain intensity. Patients’ blinding was achieved by using electrodes and being stimulated with the minimal current that the patient could feel in the same therapeutic setting, and the electrodes were not removed for 30 minutes. Besides, as preoperative TEAS combined with intraoperative TEAS led to reduced pain intensity at postoperative 6 and 24 hours compared with preoperative TEAS alone, we believe that patients’ blinding was successful. Moreover, TEAS was achieved by using a formulaic procedure that was generated using the same apparatus, which improves the homogeneity of the methodology and avoids operator-dependent bias.

In this study, TEAS did not affect several secondary outcomes, including requirement for intraoperative anesthetics, postoperative nausea, vomiting, gastrointestinal dysfunction, extubation time, and length of stay in PACU. Further research is needed to investigate the effects of TEAS on these outcomes.

The limitations of this study should also be noted. First, although previous study reported that 5 minutes stimulation has no effect on postoperative pain,³⁵ we can not absolutely exclude a possible analgesia effect of the sham TEAS. And the lack of a negative-control patient cohort does not exclude a potential placebo effect of sham acupuncture that may partially attenuate the analgesic effect seen in the TEAS groups, partially leading to discrepancy between our and previous results.³⁶ However, since 2 sessions of TEAS reduced postoperative pain intensity compared with both sham TEAS and preoperative TEAS alone, and the difference between sham TEAS and preoperative TEAS alone was not obvious, we conclude that the analgesic effect of preoperative TEAS alone is not significant enough for clinical importance. Second, 82.3% of the participants were women because selected surgeries included laparoscopic cholecystectomy and gynecological surgery. It is well known that women generally report pain at lower stimulus intensity and have a lower tolerance to pain as compared with men and that the individual variation of the electrical pain thresholds corresponding to transcutaneous electrical stimulation was greater in women than in men.^{47–49} Thus, this is a single center study in a strictly defined patient population, which may be interpreted with care. Moreover, as acupuncture and related

technique are traditional practices in Chinese culture, the psychological effect of higher acceptance might create a higher satisfactory result than in other ethnic population.

We conclude that combined preoperative TEAS with intraoperative or postoperative TEAS, rather than preoperative TEAS alone, offers a safe alternative and adjunct treatment for pain relief following laparoscopic surgery.

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